

C Programming Interview Questions and Answers

1. What is the difference between a for loop and a while loop?

A for loop is generally used when the number of iterations is known. A while loop is used when the number of iterations is not known in advance and depends on a condition.

2. What is the difference between a struct and a union?

In a structure, each member has its own memory. In a union, all members share the same memory location.

3. What is a pointer? Why is it used?

A pointer is a variable that stores the address of another variable. It is used for dynamic memory allocation, passing arguments by reference, and efficient array handling.

4. What is the difference between a local variable and a static variable?

A local variable is created and destroyed every time the function is called. A static variable retains its value between function calls.

5. What is recursion?

Recursion is a technique where a function calls itself until a base condition is reached.

6. What is the difference between break and continue?

break exits the loop completely. continue skips the current iteration and continues with the next iteration.

7. What is the difference between ++i and i++?

++i increments first, then uses the value. i++ uses the current value first, then increments.

8. What is a segmentation fault?

A segmentation fault occurs when a program tries to access invalid or unauthorized memory, such as dereferencing a NULL or uninitialized pointer.

9. What is the difference between an array and a pointer?

An array stores a fixed number of elements. A pointer stores the address of a variable or memory location.

10. What is the output? `int a=10; int b=a++; printf("%d %d",a,b);`

Output: 11 10. Explanation: a++ assigns the current value to b first, then increments a.